Homework 10

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Complete the socket examples we started in class.

2. Both `server1` and `server2` allow for communication between two terminals on a lab machine. However, one uses a Unix domain socket and the other an Internet domain socket. Explain the difference in how the kernel identifies these two endpoints.

3. Look at the source code for `client1` and `client2`. How does the address passed to the `connect()` system call differ between a Unix socket and an Internet socket?

4. While the server is running on `syslab01` and the client on `netlab05`, identify the four components of the socket pair that uniquely identify this specific connection.

5. In Example 1, you must run the server first. In Example 4, what happens if you run the client before the server? Relate this to the “handshaking” or “connectionless” nature of the protocols.

6. If a packet is dropped while running Example 4, does the UDP protocol attempt to retransmit it? Why is this acceptable for real-time applications like audio/video streaming?

7. If you and another student both try to run `server2` on the same lab machine using the same port, what error will the second student receive? How does the 16-bit port number solve the problem of multiple processes running on a single host ?

8. When you run `client3`, the kernel assigns it an ephemeral port. Does this port number stay the same every time you run the client, or does it change? Why?

- **Welcome: Sean**
  - LogOut

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